温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Data Visualization II

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Week 6-2, 09 Oct 2025

Outline

- Tables
  - When and how to use effectively
  - Crosstabulation / Contingency table
- Charts
  - Overview of common charts types
  - Some examples
- Demo
  - Create and refine a **line chart** using *matplotlib*
- Recap on Assignment 1

Data Visualization – Tables

- When to Use Tables
  - To provide specific numerical values
  - To allow precise comparisons between values
  - When values have different units or very different magnitudes

Example: Table of monthly costs and revenues

	Month						
	Jan	**Feb**	**Mar**	**Apr**	**May**	**June**	**Total**
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,125	67,125	48,178	51,785	55,678	353,015

Source: Camm Ch3

Data Visualization – Tables

- Table Design Tips
  - Minimize clutter → keep the data-ink ratio high
  - Use lines sparingly → only to separate labels, data, and calculated fields
  - Align consistently
    - Labels: left-aligned
    - Values: right-aligned
    - Vertical labels: centered

Design A:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design B:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design C:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design D:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Source: Camm Ch3

Data Visualization – Tables

Suppose that you wish to display data on revenues, costs, and head count for each month.

- Costs and revenues are measured in **dollars**, but head count is measured in **number of employees**.
- Because of the different levels of magnitudes, a table is preferred over a chart.

	Month						
	Jan	**Feb**	**Mar**	**Apr**	**May**	**June**	**Total**
Head count	8	9	10	9	9	9	
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,125	67,125	48,178	51,785	55,678	353,015

Source: Camm Ch3

Data Visualization – Crosstabulation (Cross-tab / Contingency)

- A method to summarize the relationship between two (or more) categorical variables
- Presents data in a **matrix** (table) format
  - rows = categories of one variable
  - columns = categories of another
- Shows **frequency counts** (or **percentages**) for each category combination
- Helps reveal patterns, relationships, or associations between variables

	Meal Price				
Quality Rating	**$10-19**	**$20-29**	**$30-39**	**$40-49**	**Total**
Good	42	40	2	0	84
Very Good	34	64	46	6	150
Excellent	2	14	28	22	300
Total	78	118	76	28	300

Source: Camm Ch3

Data Visualization – Charts

Basic Principles of Chart Selection

- Define the purpose
  - Clarify the goal: *comparison, distribution, relationship, trend, etc.*
- Know your data
  - Data type: *categorical, ordinal, numerical, time series*
  - Data dimensions
- Consider your audience
  - Match the chart design to their background and familiarity with the data
- Keep it simple
  - Choose the **clearest, most direct** chart type to convey information
- Avoid misleading visuals
  - Prevent distortion (e.g., truncated axes, inappropriate proportions)

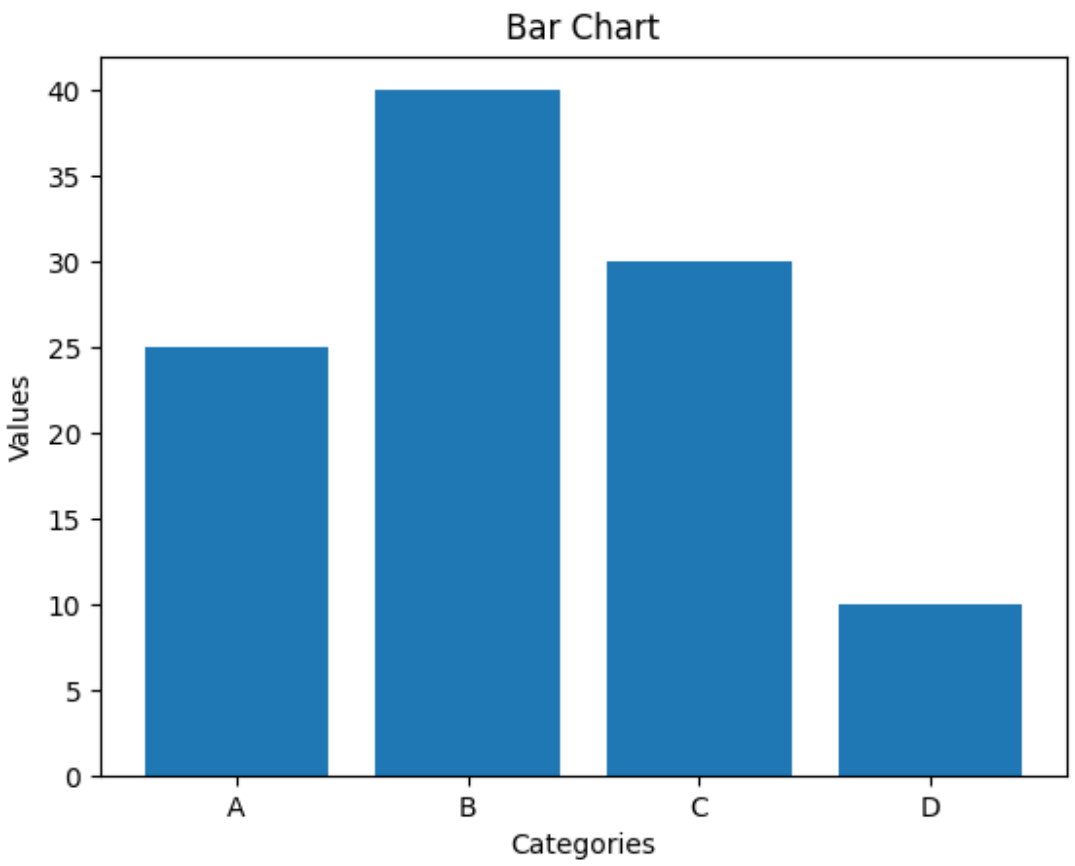
Data Visualization – Charts

Recommended Charts by Analysis Purpose

Analysis Purpose	Recommended Chart Types
Compare values across categories	Column / bar / radar chart
Show **part-to-whole** relationships	Pie / donut / stacked column chart, tree map
Analyze data **distribution**	Histogram, box /scatter / density / bubble plot, heat map
Display **trend**s over time	Line / multi-line / area / stacked-area / spline / candlestick chart
Show **relationships & correlations**	Scatter plot, bubble chart, hear map, correlation matrix
Display **geographic** data	Map, heat map, point map, flow map
Display **hierarchical structure**	Tree map, sunburst chart, Sankey diagram
Show **network & connections**	Network graph, chord diagram, force-directed graph

- *A chart could be used for different purposes.*
- *Beyond the above basics, many other chart types and variations exist.*

Data Visualization – Bar Chart

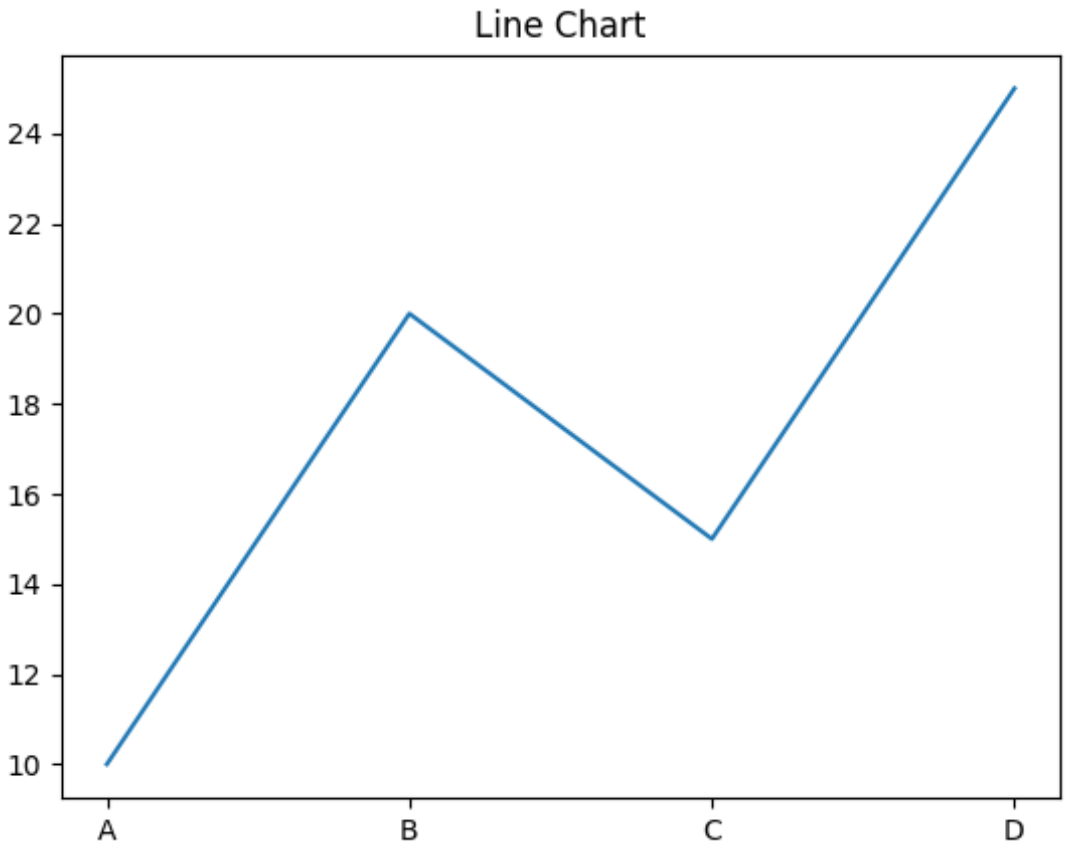


When to Use:

- To compare different categories
- To rank values from highest to lowest
- To show relationships between multiple variables

Types of Data Visualization Charts: From Basic to Advanced

Data Visualization – Line Chart

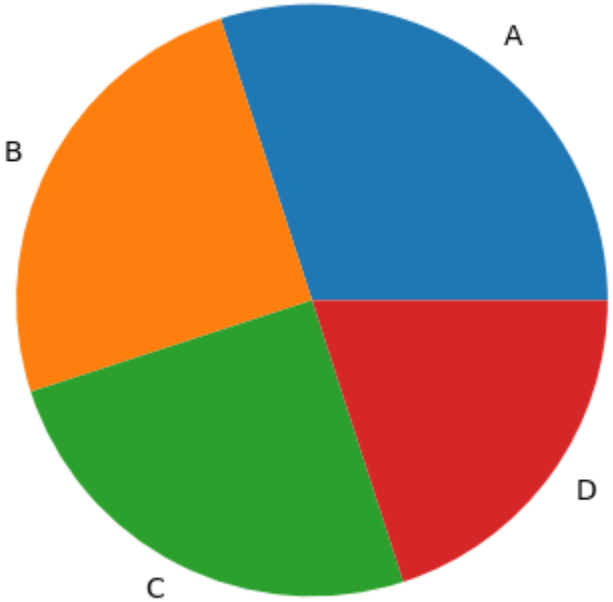


When to Use:

- To track changes over time
- To compare trends across multiple data series
- For time series analysis

Types of Data Visualization Charts: From Basic to Advanced

Data Visualization – Pie Chart

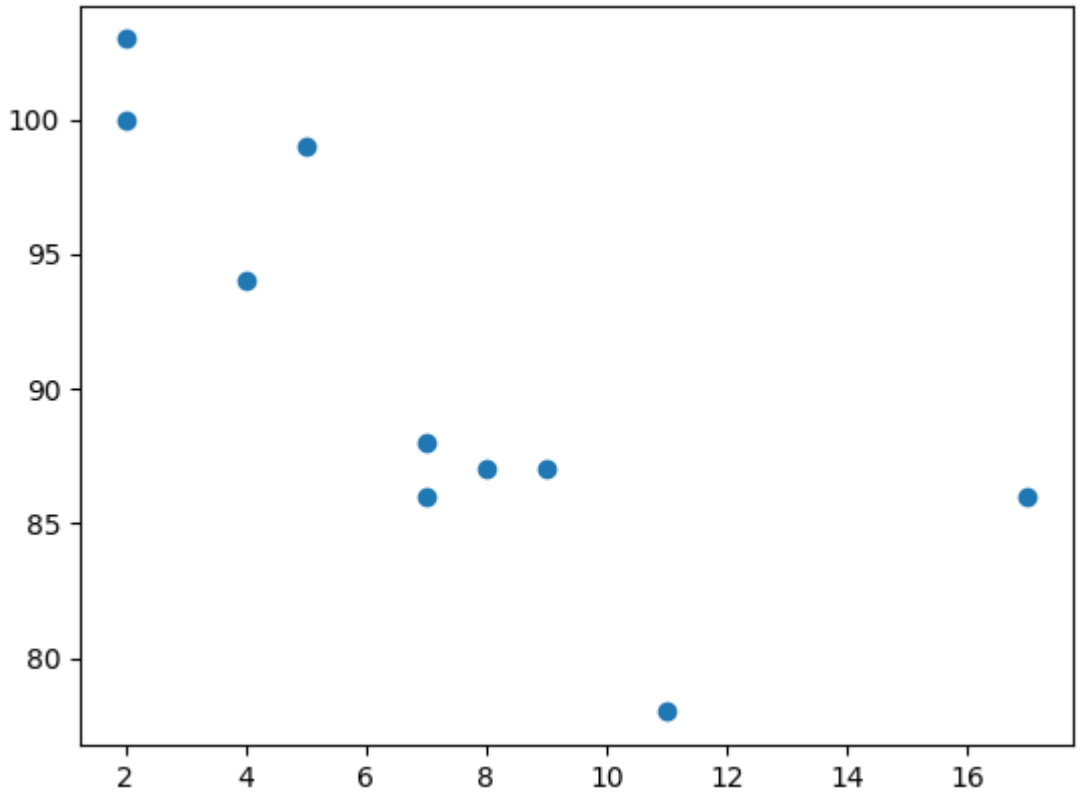


When to Use:

- To show how different parts contribute to a whole
- To highlight a dominant category

Types of Data Visualization Charts: From Basic to Advanced

Data Visualization – Scatter Chart (Plot)

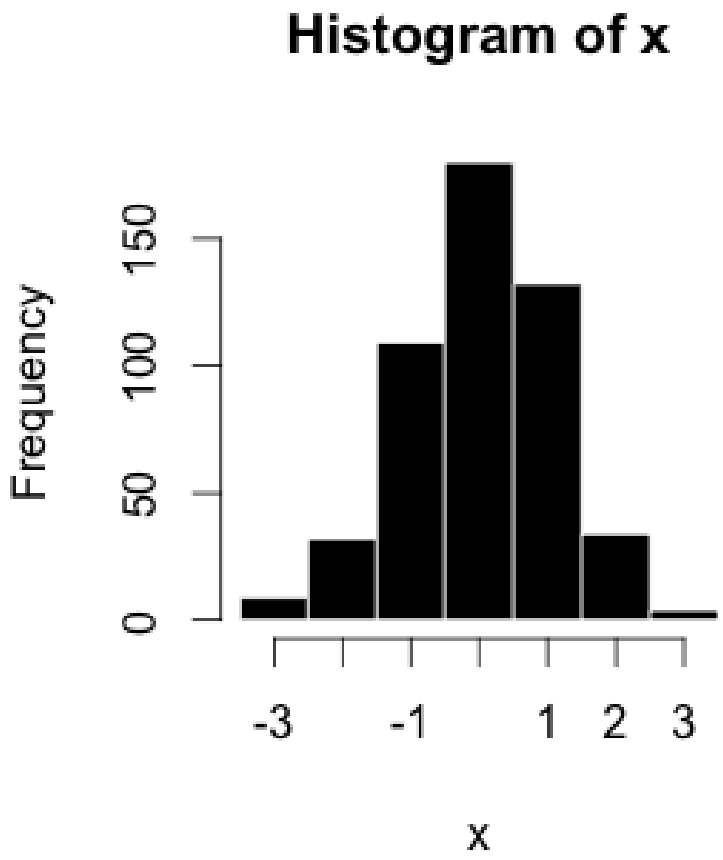


When to Use:

- To observe relationships between two variables
- To detect patterns, clusters or outliers in data

Types of Data Visualization Charts: From Basic to Advanced

Data Visualization – Histogram



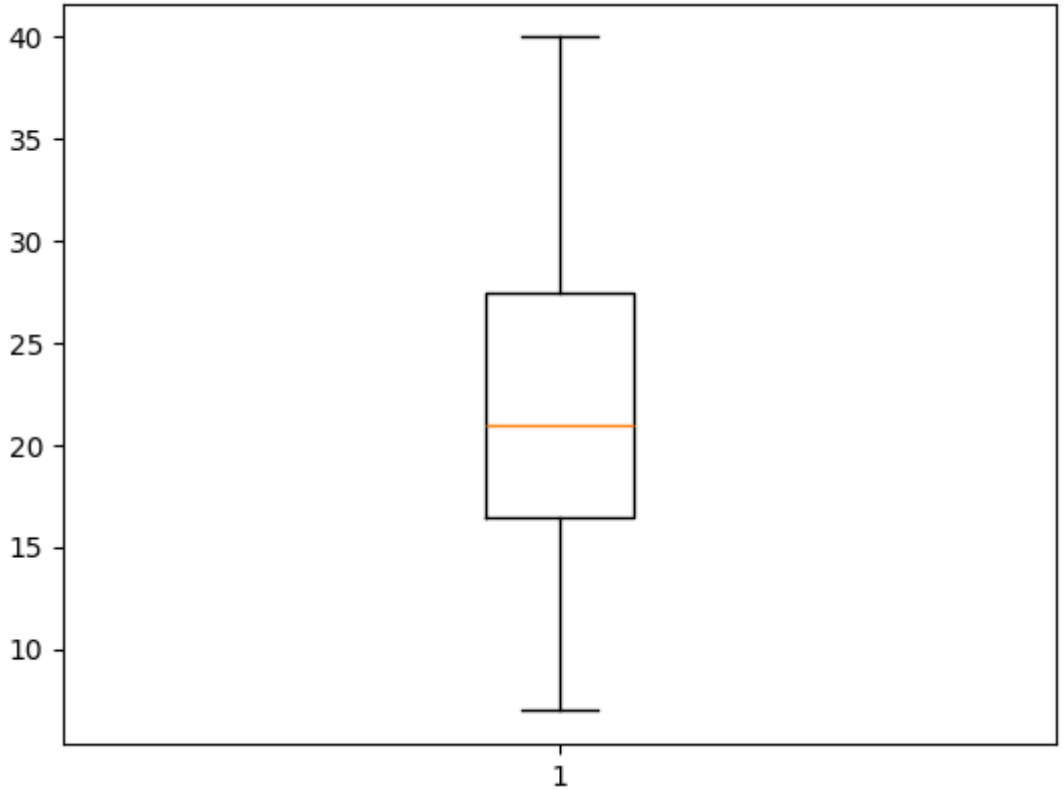
When to Use:

- To visualize the distribution of numerical data
- To explore patterns, trends and outliers

Question: What is the difference between bar chart and histogram?

Types of Data Visualization Charts: From Basic to Advanced

Data Visualization – Box Plot (Box-and-Whisker Plot)

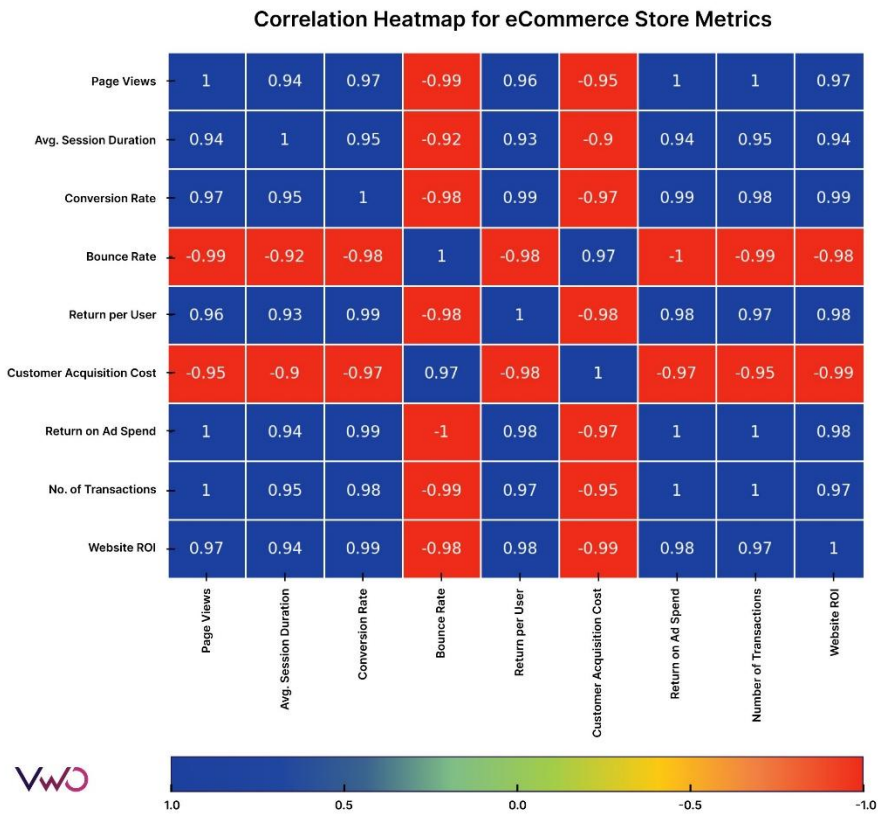


When to Use:

- To detect outliers in data
- To compare distributions across groups
- To visualize spread and variability

Types of Data Visualization Charts: From Basic to Advanced

Data Visualization – Heat Map



https://vwo.com/website-heatmap/how-to-read-heatmap/

When to Use:

- To identify clusters or groupings in data
- To visualize correlations between variables
- For risk analysis in fields like finance or network security

Types of Data Visualization Charts: From Basic to Advanced

Demonstration – Line Chart using Python

- Modules
  - Matplotlib → the *foundation* plotting library in Python
  - Seaborn → a *high-level* interface built on top of Matplotlib
- Demo Overview
  - We will create and refine a **line chart** using *matplotlib*.
  - Please download:
    - sales.csv dataset
    - demo_line_chart.ipynb instruction file
- Two Ways to Create Plots in Matplotlib
  - Pyplot (state-based) interface → using plt.plot()
  - Object-oriented (OO) interface → using fig and ax objects
- Improve Readability
  - **Every element** of a figure can be customized (e.g., title, labels, legend, etc.)
  - Remember our discussion from last class on "*what makes a good data visualization*"

Recap on Assignment 1

- Use of AI
  - I can often tell who used AI and who did not.
  - But it is **difficult, yet impossible**, to tell how it was used.
    - 100% AI-generated
    - AI-generated + human edits
    - Human-written + AI polishing
    - Human x AI co-creation
  - I will not *detect, judge, or penalize* AI use due to false positive.
  - I am not against AI use — it is important to learn to use it responsibly and ethically.
  - What matters most is how you use AI.
  - Future assignments will be designed to discourage **"easy"** AI-generated answers.
  - AI use is **prohibited** in the midterm, final exam, and interview.

Recap on Assignment 1

- Common Issues
  - Grammar: Fix blue/red wavy underlines in MS Word
  - Structure: Use **bullet points**, ensure a clear and logical flow
  - **Read questions carefully**:
    - (1a) → Need both correct classification and justification
    - (2a) → Include descriptive, predictive, and prescriptive examples
- Submission & Feedback
  - Late submission: **50% deduction**, based on Beijing Time
  - Grades & answers: Posted in Canvas
    - Learn clear structure and concise justification (no need to over-explain)
  - Grades review policy
    - Questions? See me within one week after grades are posted
    - After that, grades are final and non-negotiable

Recap on Assignment 1

Good vs Not-So-Good Example

*(a) What types of data (categorical, ordinal, interval, or ratio) would each of the survey items represent and **why**?*

(a) ①Gender-Categorical: Gender is textual and labeled data that can be used to classify data into different groups, without order.
②Age-Ratio: Age is a continuous value with an absolute zero point.
③Ethnicity-Categorical: It is used to identify the affiliated, and there is no sequence among these categories.
④Length of residency in Wenzhou -Ratio: It is measured in years, months, etc. It is a continuous quantity and has a real zero point.
⑤Overall satisfaction with shopping mall services-Ordinal: It is an evaluation index scored from 1 to 5 points, with a clear order, for example, 5 points is better than 4 points.
⑥Quality of food-Ordinal: It is an evaluation index scored from 1 to 5, which is orderly.

a):Gender: Categorical data.Because gender can be classified into different categorie

Age: Ratio data.Because age has a starting point of zero.

Ethnicity: Categorical data.Because races can be classified into different categories

Length of residency in Wenzhou: Ratio data.Because the duration of residence is calculated from zero.

Overall satisfaction with shopping mall services: Ordinal data.Because it has an ordered ranking.

Quality of food: Ordinal data.Because it has an ordered ranking.

Recap on Assignment 1

Good vs Not-So-Good Example

(b) Because the data type determines the statistical methods, chart formats, data processing skills you can use. If the data type is defined incorrectly, it may lead to wrong analysis methods and even incorrect conclusions, thereby misleading business decisions.

EX1: If "male" is coded as 1 and "female" as 2 in a survey, treating gender as a numerical variable. If we want to calculate the average gender, it could result in a value like 1.5. This has no statistical meaning, as gender is a categorical variable with no numerical and ordinal relationship. Moreover, if gender is included in a regression model as a numerical variable, the model may wrongly interpret it that a higher gender value leads to a stronger effect, which results in a misleading interpretation.

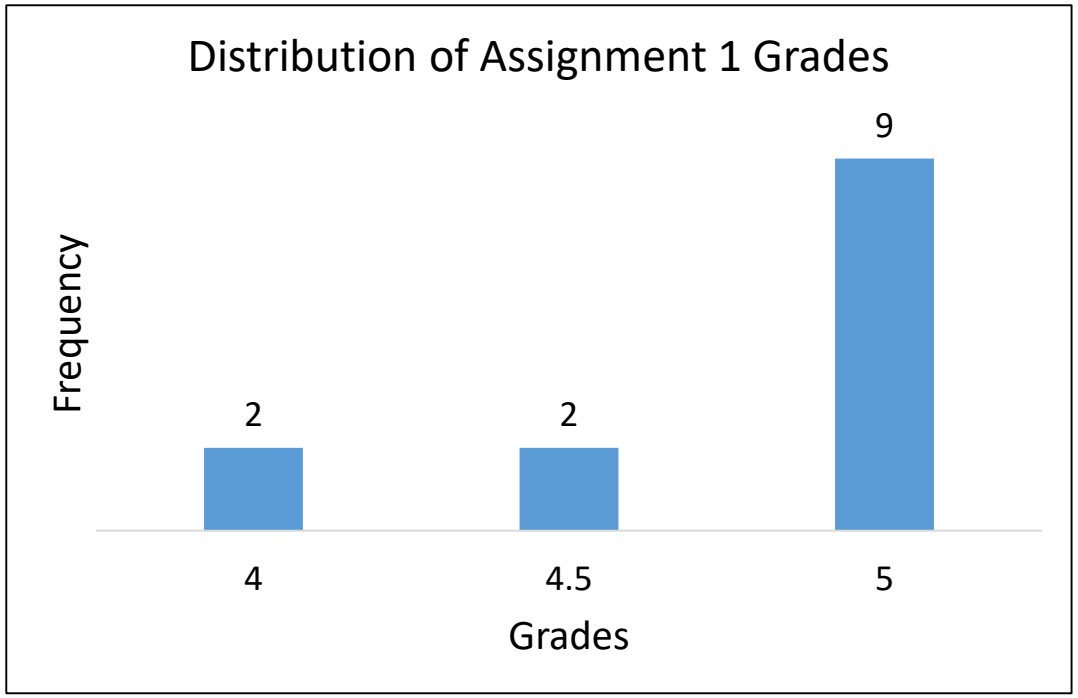
EX2: If "length of residence" is simply categorized into groups like short-term, medium-term and long-term before analysis and treated as a categorical variable. This may capture general trends but sacrifices the original numerical precision and continuity. As a result, the analysis is limited to comparing average satisfaction across groups, making it impossible to explore more detailed relationships such as whether each additional year of residence significantly increases satisfaction.

(b) Why is it important to define the data type in your data file correctly before conducting data analyses? Use one or two examples to explain your answer.

b)

a. For example, if we want to define age index. If we consider it is categorical data. When we analyze the data. We will find that we can only find specific ages' features instead of we can also find some mean value in ages and so on.

Recap on Assignment 1



First Assignment

- Treat it as an experiment — a chance to learn and adjust
- It is just a small part of your overall grade, will not significantly affect your final score.